Frames and matrix designs

Complex equiangular tight frames with twice the dimension

Difficulty: extreme **Importance:** broadly interesting

Status: source-backed open problem; no resolution found in the bounded literature check.

Conjecture. For every integer $d \geq 2$, there exist $2d$ vectors $v_1, \dots, v_{2d} \in \mathbb{C}^d$ satisfying

$$\|v_i\|_2 = 1, \qquad \sum_{i=1}^{2d} v_i v_i^* = 2I_d, \qquad |v_i^* v_j|^2 = \frac{1}{2d-1} \quad (i \neq j).$$

Thus the columns of $V = [v_1 \ \cdots \ v_{2d}]$ form a unit-norm equiangular tight frame. The target requires every dimension, rather than only an infinite family.

These are optimal line configurations at redundancy two: their coherence attains the Welch bound. They provide well-balanced matrix designs for sensing and reconstruction. This is distinct from the d^2-vector SIC problem in FR-07.

References

1. A. Glazyrin, *New constructions of optimal arrangements of* $2d$ *lines in* $\mathbb{C}^d$, August 2026, Conjecture 1 (attributed to Fallon and Iverson), and Sections 2–5 for constructions. Paper.

Status check — 2026-09-08

Glazyrin explicitly retains the all-dimension conjecture while extending the known constructions. Searches for the Fallon–Iverson conjecture and later redundancy-two ETF results found no full resolution. Finite dimension verification and the new tensor/power constructions do not cover the universal statement.